18.3 Gradient fields and path-independent fields, part 2 p.1 of 1

Let $\vec{F} = \langle F_1, F_2 \rangle$ be a smooth vector field. Evaluating

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}$$

at a point (x,y) gives the scalar curl (or circulation) at that point. Counterclockwise rotation is counted positive, while clockwise rotation is counted negative.

Fact: If $\vec{F} = \nabla f$ so that $F_1 = \frac{\partial f}{\partial x}$ and $F_2 = \frac{\partial f}{\partial y}$, then

$$\frac{\partial F_2}{\partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial F_1}{\partial y}$$

Thus if $\vec{F}$ is conservative, it has zero curl.

Summary:

$$\begin{tikzcd}
\boxed{\vec{F} \text{ is conservative}} \arrow[r] \arrow[d, leftrightarrow] & \boxed{\vec{F} \text{ has zero curl}} \arrow[l, dashed, "\text{no holes}"] \arrow[d, dashed, "\text{no holes}"] \\
\boxed{\vec{F} \text{ is path-independent}} \arrow[r, leftrightarrow] & \boxed{\oint_C \vec{F} \cdot d\vec{r} = 0 \text{ for all closed curves } C} \arrow[u]
\end{tikzcd}$$

- $\vec{F}$ being path-independent means $\int_C \vec{F} \cdot dr$ only depends on the end points of C.
- $\oint_C \vec{F} \cdot d\vec{r}$ is the same thing as $\int_C \vec{F} \cdot d\vec{r}$, but that C is closed, meaning it begins and ends at the same point.
- No holes means $\vec{F}$ has no holes in its domain.

→ Do all problems.